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BoardsDefTopic

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DC30/75 Instrument Boards

DC30/75 Instrument Boards

The DC30 and DC75 instrument boards provide the hardware that supports the following DC instruments:

- DC30
  - DCVI channels
  - DC Differential Voltmeters
  - DC Time Measurement Units
- DC75
  - DCVI channels
  - DC Differential Voltmeters
  - DC Time Measurement Units

This section provides an overview of each board in the following sections:

- DC30 Instrument Board
- DC75 Instrument Board

For more information on DC instruments, see:

- About the DCVI
- About the DCDiffMeter
- About the DCTime

For information on DC hardware specifications, see:

- DC30 Specifications
- DC75 Specifications

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DC30 Instrument Board

DC30 Instrument Board

The DC30 DCVI is a multichannel DC instrument board designed to source and sink current and voltage, measure current and voltage, and make differential and time measurements, allowing multiple test applications on a DUT.

The DC30 instrument board provides hardware that supports the following DC instruments:

- 20 DCVI channels
- 2 DC Differential Voltmeters (DCDiffMeters)
- 2 DC Time Measurement Units (DCTimes)

The DC30 instrument board includes the following features:

- Selectable forcing mode: Force voltage, force current, high impedance, high regulation
- Merged channel configurations
- Voltage/current meter per channel
- Selectable forcing bandwidth
- 4 asynchronous trigger lines

- 4 Device Ground Sense lines
- 10 DIB access pins
- DCTime access through DCVI sense pins or dedicated high performance input

This section provides a general overview of the board and includes the following topics:

- Simulated Configuration Files
- DC30 Channel Connections
- Connecting to a Live DUT from Force (V or I) Mode
- Device Ground Sense
- Using DC30 in Concurrent Testing
- DC30 Shutdowns
- DC30 Software Licensing Requirements

For more information on DC30 instruments, see:

- About the DCDiffMeter
- About the DCTime
- About the DCVI
- DC Debug Displays

Related Topics

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DC30 Instrument Board : Simulated Configuration Files

Simulated Configuration Files

You can use the DC30 offline by forcing IG-XL to map the DC30. Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- \tester folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- \bin folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

The following SimulatedConfig.txt file maps the DC30 board in slot 13:

```
<Version>
1
</Version>

<TEST_HEAD TH 1>
#slot[.subslot]   Type              idprom (type serial rev company)

4.0               HSD-M             979-219-00 0000000 0000-A 0000
                  HSD-M             956-361-00 0000000 0000-A 0000

12.0              HexVS             974-294-00 0000000 0000-A 0000
                  HexVS             956-388-00 0000000 0000-A 0000

13.0              DC-30             974-217-30 0000000 0000-A 0000
```

DC-30 956-085-10 0000000 0000-A 0000

21.0 VSM 974-296-00 0000000 0000-A 0000
VSM 956-361-00 0000000 0000-A 0000

24.0 SupportBoard 974-220-12 0000000 0000-A 0000
SupportBoard 956-354-00 0000000 0000-A 0000

62.0 DistributionBoard 956-355-00 0000000 0000-A 0000
DistributionBoard 956-355-00 0000000 0000-A 0000

</TEST_HEAD TH 1>

To use the DC30 offline, the SimulatedConfig.txt file must include every board needed in the IG-XL workbook. The SimulatedConfig.txt file represents the state of the offline tester. To use the DC30 online, you need not edit any files. IG-XL automatically maps all boards in the tester.

For more information, see Tester Configuration.

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DC30 Instrument Board : DC30 Channel Connections

DC30 Channel Connections

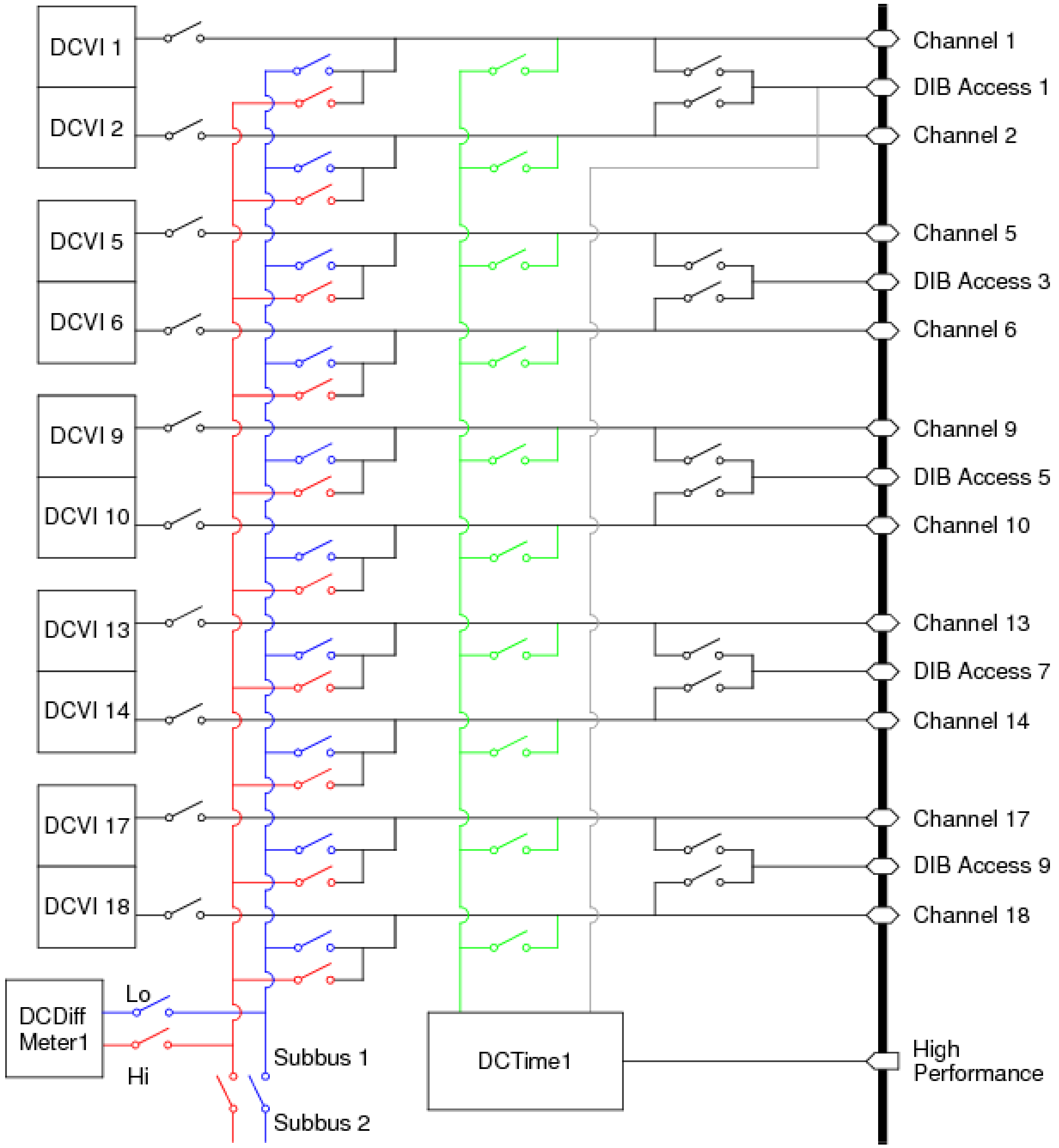
Each DC30 has 20 DCVI channels, 2 DC Differential Voltmeters, and 2 DC Time Measurement Units that share connections to the DUT.

Each DC Differential Voltmeter is located on one of two subbusses allowing direct connections between 10 channels and indirect connections to 10 channels.

Each DC Time Measurement Unit shares channels with a designated set of 10 DCVIs. This figure shows the connections shared between 10 DCVI channels, one DC Differential Voltmeter, and one DC Time Measurement Unit. Connections for the channel are simplified showing the force, sense, and guard as one line. Plan DIB layout carefully.

☑ **Note:**

Avoid connecting a DCVI to a live (powered) DUT pin. Doing so may potentially damage the DUT or adversely affect instrument reliability mainly due to the relays being hot switched.

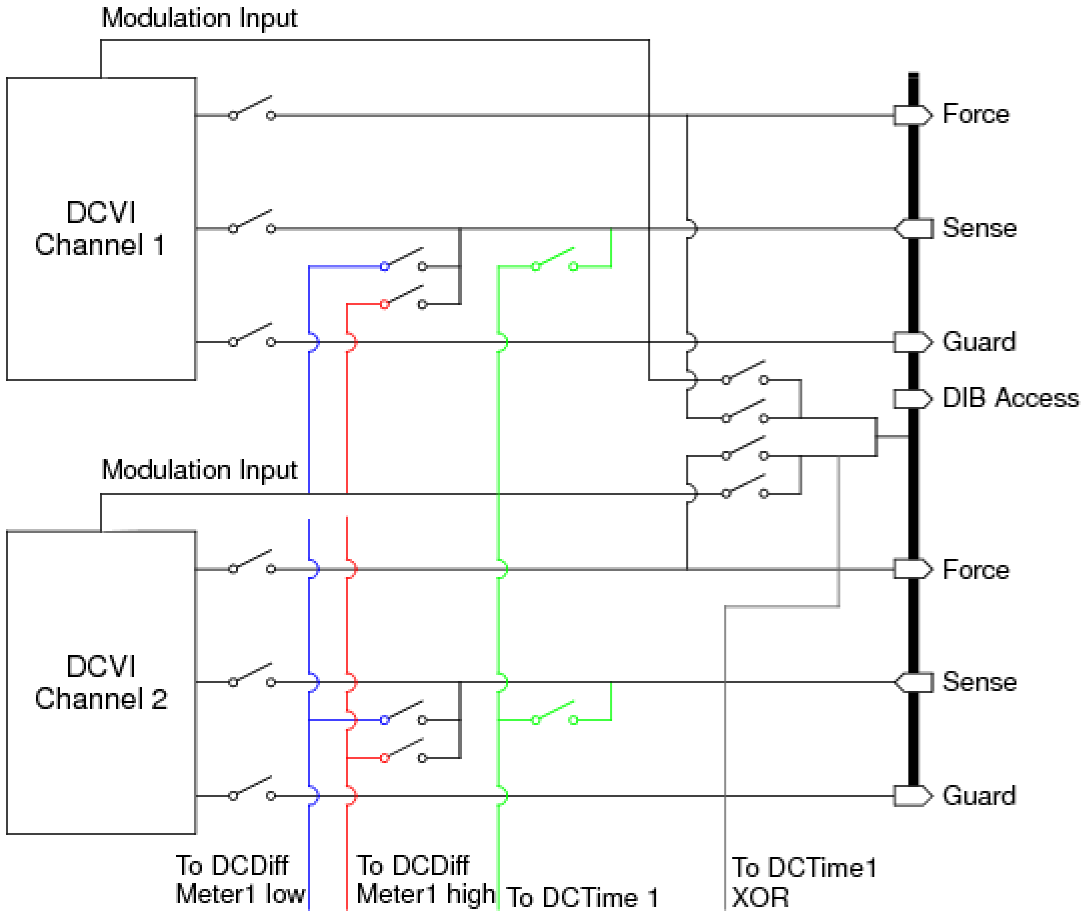


Connections Diagram for 10 DC30 Channels

You can make the following connections:

- DCVI to the DIB through force and sense lines
- DCDiffMeter from the DIB through sense lines
- DCTime from the DIB through the high performance lines
- DCTime from the DIB through sense lines

The following figure shows the force, sense, and guard lines independently to show the connections between the three on-board instruments in greater detail.



Detailed Connections for Two DC30 Channels

For more information, see:

- DC30 DGS Lines
- Connecting a DCDiffMeter
- Connecting a DCTime
- Connecting a DCVI

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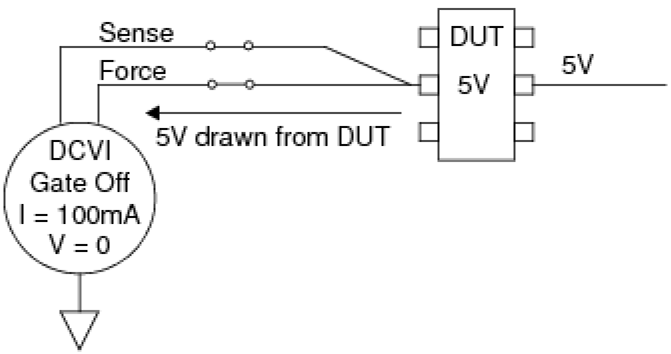
DC30 Instrument Board : Connecting to a Live DUT from Force (V or I) Mode

Connecting to a Live DUT from Force (V or I) Mode

If you must connect to a live DUT from Force (V or I) Mode, use the following best practices.

Potential problems of connecting to a live DUT are shown in the figure.



Connecting to a Live DUT Pin

The following conditions exist in the setup shown in this figure:

- Gate is Off
- Voltage = 0V (Default Gate Off setting)
- Current = 100mA
- DUT is at a 5V potential

Before connecting a DCVI to the DUT, a 5V potential exists between the DCVI and the DUT. As the DCVI relay closes, current will flow between the DCVI and the DUT. This current can damage the relay.

Take the following steps before connecting to minimize the impact of hot switching DCVI relays:

- Program the current range to the lowest instrument range.
- Program the DCVI to the slowest bandwidth setting.
- Program the DCVI voltage as close as possible to the DUT pin.
- Program the Gate to On.
- Program a wait settling time of about 2ms.
- Connect the DCVI.
- Wait the relay settling time.
- Reprogram the voltage and current values required for the test.

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DC30 Instrument Board : Device Ground Sense

Device Ground Sense

Device Ground Sense (DGS) is a sense signal that tracks the voltage of ground on the DUT and is used as a correction by the instrument to accurately force or measure voltages relative to the ground potential of the device. DGS must be connected to a single point on the DIB. The optimum place to connect DGS is at the device ground pin.

DGS connections are made by IG-XL when the instrument is connected to the DUT. Any used DGS must be grounded on the DIB or a DGS alarm can occur.

DC30 DGS Lines

The DC30 Instrument board has four DGS lines that each DCVI and DCTime connect to. You can specify DGS connections in a channel map. Otherwise, each channel defaults to a certain DGS line as shown in the table. Take care to keep track of the connections to DGS lines. The best way to do this is by specifying the DGS in the channel map.

DC30 Default DGS Lines

	DCVI		DCTime	
Channel	Inst.	DGS	Inst.	DGS
1	1	1	1	1
2	2	1	1	1
3	3	1	2	2
4	4	1	2	2
5	5	1	1	1
6	6	2	1	1

7	7	2	2	2
8	8	2	2	2
9	9	2	1	1
10	10	2	1	1
11	11	3	2	2
12	12	3	2	2
13	13	3	1	1
14	14	3	1	1
15	15	3	2	2
16	16	4	2	2
17	17	4	1	1
18	18	4	1	1
19	19	4	2	2
20	20	4	2	2

DCDiffMeter DGS

The DCDiffMeter is only connected to a DGS line when specified in a channel map for use on the Low Side of the DCDiffMeter. For more information, see Programming the Channel Map in the DCDiffMeter section.

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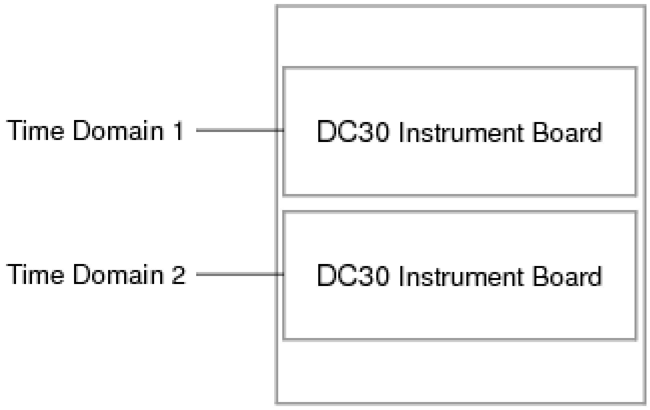
DC30 Instrument Board : Using DC30 in Concurrent Testing

Using DC30 in Concurrent Testing

To support concurrent testing, you can assign each DC30 instrument to a unique time domain. Concurrent features for DC30 include:

- Each DC30 board can be configured for one time domain
- Each DC30 instrument can be controlled by a unique time domain using Sync-Link. Features include:
  - Independent access to Sync-Link
  - Independent sample clock programming, one sample rate per channel
  - Independent clock synchronization with other DC30 instruments or with digital clocks
  - Independent control by digital pattern
  - Independent capture memory and triggers
- Multiple instruments can be grouped into a common time domain
- DC30 is restricted to using the Global flow domain and synchronous pattern starts

The following figure shows a configuration with each DC30 instrument assigned to its own time domain.



DC30 Concurrent Testing Configuration Example, Two Time Domains

For more information, see About Concurrent Test.

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DC30 Instrument Board : DC30 Shutdowns

DC30 Shutdowns

A shutdown indicates that a condition exists that could potentially damage an instrument. Repeatedly invoking a shutdown on any instrument can adversely affect the reliability of all instruments in the test system.

The DC30 responds to or invokes a system shutdown and then executes a shutdown sequence. This is a sequence of events that a DC30 goes through to protect itself from damage and to remove potentially dangerous conditions. Whenever an instrument shutdown occurs, each instrument that responds to the shutdown also executes a shutdown sequence specific to that instrument.

The following topics are included:

- DCVI Shutdown Conditions
- DCTime Shutdown Conditions
- Clearing a Shutdown Condition

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DC30 Instrument Board : DC30 Shutdowns : DCVI Shutdown Conditions

DCVI Shutdown Conditions

A DCVI invokes a shutdown when the following condition exists:

- Over Voltage

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DC30 Instrument Board : DC30 Shutdowns : Over Voltage

Over Voltage

Detects if the DCVI output is being driven above +40V or below -40V.

Shutdown Message

The following is the datalog shutdown message:

{-1} : alarm [“PinName", SiteNum] DVI:0021 : OVER-VOLTAGE SHUTDOWN. Output voltage has exceeded the programmed compliance range.

where:

PinName	Name of the pin the shutdown occurred on.
SiteNum	Site number the shutdown occurred on.

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DC30 Instrument Board : DC30 Shutdowns : DCTime Shutdown Conditions

DCTime Shutdown Conditions

A DCTime invokes a system shutdown when the following condition exists:

- Over Voltage

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DC30 Instrument Board : DC30 Shutdowns : Over Voltage

Over Voltage

The DCTime input voltage has exceeded the programmed front end voltage range.

Shutdown Message

The following are the datalog shutdown messages:

{-1} : alarm ["PinName", Site #] TMU:0015 : POSITIVE OVER VOLTAGE SHUTDOWN. Input voltage has exceeded the DCTime front end voltage range.

{-1} : alarm ["PinName", Site #] TMU:0016 : NEGATIVE OVER VOLTAGE SHUTDOWN. Input voltage has exceeded the DCTime front end voltage range.

where:

PinName	Name of the pin the shutdown occurred on.
SiteNum	Site number the shutdown occurred on.

Common Causes

An over voltage shutdown can be caused by the following:

- Defective device
- Incorrect front end range selection

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DC30 Instrument Board : DC30 Shutdowns : Clearing a Shutdown Condition

Clearing a Shutdown Condition

To clear a system shutdown, IG-XL must call an initialize routine. This means that:

- The test program in which the shutdown occurred must be stopped.
- DataTool must be stopped and started.
- The job must be revalidated.

Following a shutdown clear, make sure to understand and resolve the cause of the shutdown before rerunning the program.

For more information, see Shutdowns.

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DC30 Instrument Board : DC30 Software Licensing Requirements

DC30 Software Licensing Requirements

Operation of the DC30 does not require a software license.

For more information, see IG-XL Licensing in IG-XL Administration help.

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DC75 Instrument Board

DC75 Instrument Board

The DC75 DCVI is a multichannel DC instrument board designed to source and sink current and voltage, measure current and voltage, and make differential and time measurements, allowing multiple test applications on a DUT.

The DC75 instrument board provides the hardware that supports the following DC instruments:

- 4 DCVI channels
- 2 DC Differential Voltmeters (DCDiffMeters)
- 2 DC Time Measurement Units (DCTimes)

The DC75 instrument board includes the following features:

- Selectable forcing mode: Force voltage, force current, high impedance, high regulation
- Merged channel configurations
- Voltage/current meter per channel
- Three selectable meter filters
- Selectable forcing bandwidth

- Source and capture memory per channel
- 4 asynchronous trigger lines
- 4 Device Ground Sense lines
- 1:2 multiplexed DCVI output
- 8 DIB access pins
- DCTime access through DCVI sense pins or dedicated high performance input

This section provides a general overview of the board in the following sections and includes the following topics:

- Simulated Configuration Files
- DC75 Channel Connections
- Connecting to a Live DUT from Force (V or I) Mode
- Device Ground Sense
- Using DC75 in Concurrent Testing
- DC75 Shutdowns
- DC75 Software Licensing Requirements

For more information on DC75 instruments, see:

- About the DCDiffMeter
- About the DCTime

- About the DCVI
- DC Debug Displays

Related Topics

- DC75 Specifications
- DC75 DIB Design Guidelines

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DC75 Instrument Board : Simulated Configuration Files

Simulated Configuration Files

You can use the DC75 offline by forcing IG-XL to map the DC75. Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- \tester folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- \bin folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

The following SimulatedConfig.txt file maps the DC75 board in slot 13:

```
<Version>
1
</Version>

<TEST_HEAD TH 1>
#slot[.subslot]   Type          idprom (type serial rev company)

4.0          HSD-M          979-219-00 0000000 0000-A 0000
             HSD-M          956-361-00 0000000 0000-A 0000

12.0         HexVS          974-294-00 0000000 0000-A 0000
             HexVS          956-388-00 0000000 0000-A 0000

13.0         DC-75          974-230-00 0000000 0000-A 0000
```

DC-75 956-099-00 0000000 0000-A 0000

21.0 VSM 974-296-00 0000000 0000-A 0000
VSM 956-361-00 0000000 0000-A 0000

24.0 SupportBoard 974-220-12 0000000 0000-A 0000
SupportBoard 956-354-00 0000000 0000-A 0000

62.0 DistributionBoard 956-355-00 0000000 0000-A 0000
DistributionBoard 956-355-00 0000000 0000-A 0000

</TEST_HEAD TH 1>

To use the DC75 offline, the SimulatedConfig.txt file must include every board needed in the IG-XL workbook. The SimulatedConfig.txt file represents the state of the offline tester. To use the DC75 online, you need not edit any files. IG-XL automatically maps all boards in the tester.

For more information, see Tester Configuration.

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DC75 Instrument Board : DC75 Channel Connections

DC75 Channel Connections

Each DC75 has 4 DCVI channels, 2 DC Differential Voltmeters, and 2 DC Time Measurement Units that share connections to the DUT.

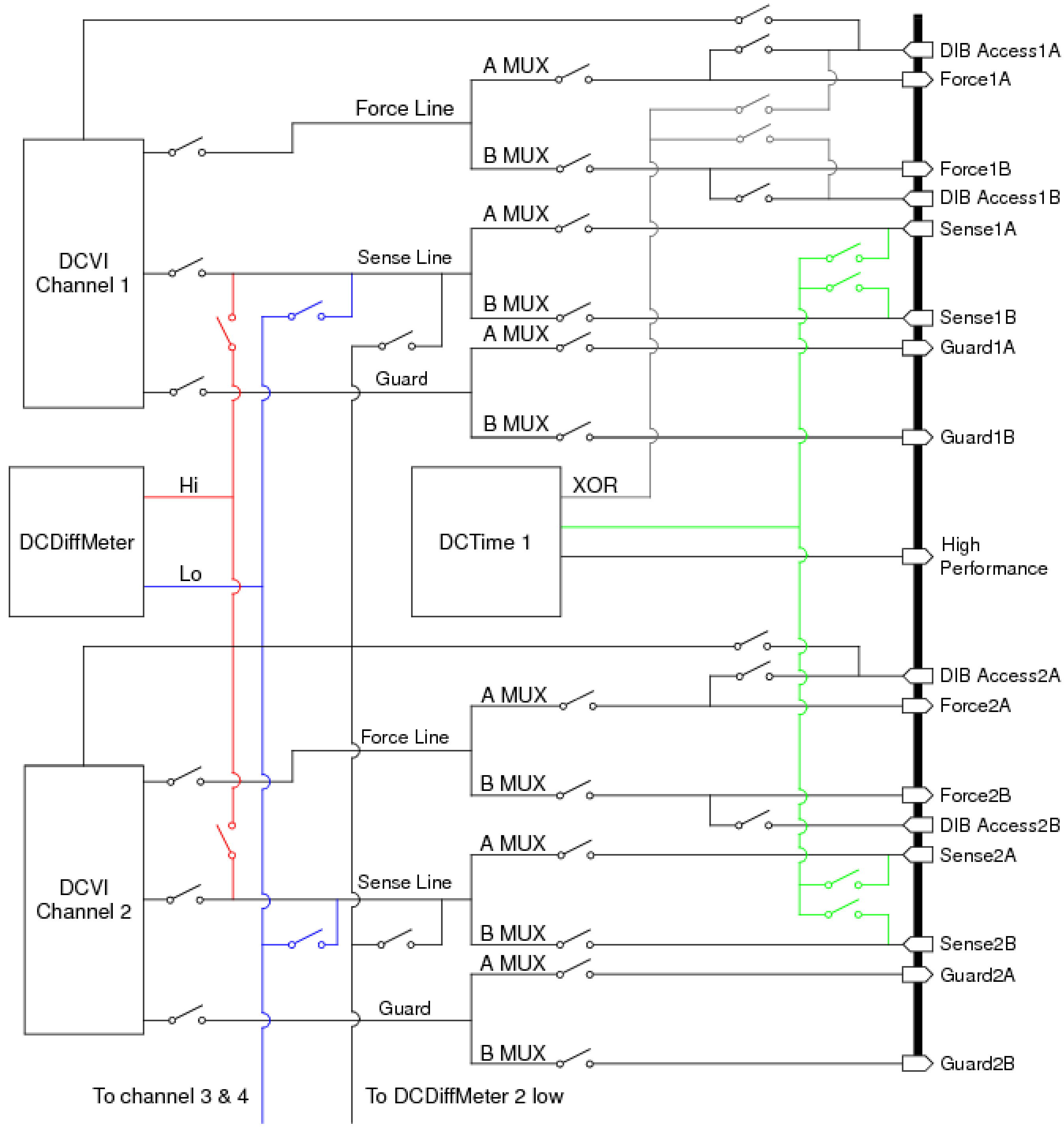
The DC75 has multiplexed outputs with each output having a DIB access line for a total of eight DIB access connections.

There are two DC Differential Voltmeters with the high side pin having access to two separate channels and the low side pin having access to all four channels. The figure shows the connection of one DC Differential Voltmeter to two channels.

Each DCTime Measurement Unit connects to a pair of channels. All instruments have access to the multiplexed outputs of the respective channels, and each DC Time Measurement Unit has access to two DIB Access pins. Plan DIB layout carefully.

☑ **Note:**

Avoid connecting a DCVI to a live (powered) DUT pin. Doing so may potentially damage the DUT or adversely affect instrument reliability mainly due to the relays being hot switched.



Connections Diagram for Two DC75 Channels

You can make the following connections:

- DCVI to the DIB through force and sense lines
- DCDiffMeter from the DIB through sense lines
- DCTime from the DIB through the high performance lines

- DCTime from the DIB through sense lines

For more information, see:

- DC75 DGS Lines
- Connecting a DCDiffMeter
- Connecting a DCTime
- Connecting a DCVI

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DC75 Instrument Board : DC75 Channel Connections : Multiplexed Connections

Multiplexed Connections

The DC75 supports internal multiplexed channels allowing more possible connections to resources without taking up DIB space. Select multiplexed channels by entering the channel DIB address for the mux you wish to connect to.

The DC75 has an A and a B side. The channel signal is passed to a mux by a series of relays that are programmed in software. Software allows independent Mux connections such as the force connected to Mux B and the sense connected to Mux A.

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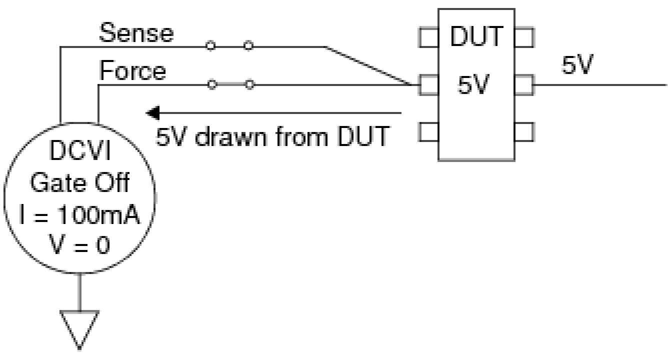
DC75 Instrument Board : Connecting to a Live DUT from Force (V or I) Mode

Connecting to a Live DUT from Force (V or I) Mode

If you must connect to a live DUT from Force (V or I) Mode, use the following best practices.

Potential problems of connecting to a live DUT are shown in the figure.



Connecting to a Live DUT Pin

The following conditions exist in the setup shown in this figure:

- Gate is Off
- Voltage = 0V (Default Gate Off setting)
- Current = 100mA
- DUT is at a 5V potential

Before connecting a DCVI to the DUT, a 5V potential exists between the DCVI and the DUT. As the DCVI relay closes, current will flow between the DCVI and the DUT. This current can damage the relay.

Take the following steps before connecting to minimize the impact of hot switching DCVI relays:

- Program the current range to the lowest instrument range.
- Program the DCVI to the slowest bandwidth setting.
- Program the DCVI voltage as close as possible to the DUT pin.
- Program the Gate to On.
- Program a wait settling time of about 2ms.
- Connect the DCVI.
- Wait the relay settling time.
- Reprogram the voltage and current values required for the test.

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DC75 Instrument Board : Device Ground Sense

Device Ground Sense

Device Ground Sense (DGS) is a sense signal that tracks the voltage of ground on the DUT and is used as a correction by the instrument to accurately force or measure voltages relative to the ground potential of the device. DGS must be connected to a single point on the DIB. The optimum place to connect DGS is at the device ground pin.

DGS connections are made by IG-XL when the instrument is connected to the DUT. Any used DGS must be grounded on the DIB or a DGS alarm can occur.

DC75 DGS Lines

The DC75 Instrument board has four DGS lines that each DCVI and DCTime connect to. You can specify DGS connections in a channel map. Otherwise, each channel defaults to a certain DGS line as shown in the table. Take care to keep track of the connections to DGS lines. The best way to do this is by specifying the DGS in the channel map.

DC75 Default DGS Lines

	DCVI		DCTime	
Channel	Inst.	DGS	Inst.	DGS
1	1	1	1	1
2	2	2	1	1
3	3	3	2	2
4	4	4	2	2

DCDiffMeter DGS

The DCDiffMeter is only connected to a DGS line when specified in a channel map for use on the Low Side of the DCDiffMeter. For more information, see Programming the Channel Map in the DCDiffMeter section.

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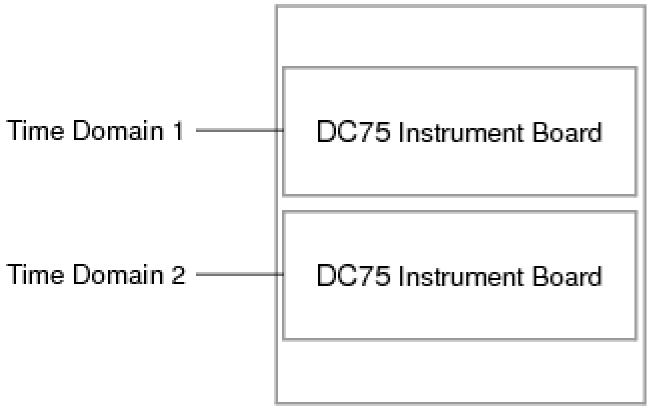
DC75 Instrument Board : Using DC75 in Concurrent Testing

Using DC75 in Concurrent Testing

To support concurrent testing, you can assign each DC75 instrument to a unique time domain. Concurrent features for DC75 include:

- Each DC75 board can be configured for one time domain
- Each DC75 instrument can be controlled by a unique time domain using Sync-Link. Features include:
  - Independent access to Sync-Link
  - Independent sample clock programming, one sample rate per channel
  - Independent clock synchronization with other DC75 instruments or with digital clocks
  - Independent control by digital pattern
  - Independent capture memory and triggers
- Multiple instruments can be grouped into a common time domain
- DC75 is restricted to using the Global flow domain and synchronous pattern starts

The following figure shows a configuration with each DC75 instrument assigned to its own time domain.



DC75 Concurrent Testing Configuration Example, 2 Time Domains

For more information, see About Concurrent Test.

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DC75 Instrument Board : DC75 Shutdowns

DC75 Shutdowns

A shutdown indicates that a condition exists that could potentially damage an instrument. Repeatedly invoking a shutdown on any instrument can adversely affect the reliability of all instruments in the test system.

The DC75 responds to or invokes a system shutdown and then executes a shutdown sequence. This is a sequence of events that a DC75 goes through to protect itself from damage and to remove potentially dangerous conditions. Whenever an instrument shutdown occurs, each instrument that responds to the shutdown also executes a shutdown sequence specific to that instrument.

The following topics are included:

- DC75 DCVI Shutdown Conditions
- DCTime Shutdown Conditions
- Clearing a Shutdown Condition

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DC75 Instrument Board : DC75 Shutdowns : DC75 DCVI Shutdown Conditions

DC75 DCVI Shutdown Conditions

A DCVI invokes a shutdown when the following conditions exist:

- Over Voltage
- Thermal

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DC75 Instrument Board : DC75 Shutdowns : Over Voltage

Over Voltage

Detects if the DCVI output is being driven above +85V or below -85V.

Shutdown Message

The following is the datalog shutdown message:

{-1} : alarm ["PinName", SiteNum] DVI:0021 : OVER-VOLTAGE SHUTDOWN. Output voltage has exceeded the programmed compliance range.

where:

PinName	Name of the pin the shutdown occurred on.
SiteNum	Site number the shutdown occurred on.

Common Causes

An over voltage shutdown can be caused by the following:

- Incorrect DCVI output mux selected
- Instrument miswired on the DIB
- Insufficient compliance settling time
- Defective device

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DC75 Instrument Board : DC75 Shutdowns : Thermal

Thermal

Indicates the instrument temperature is too high to continue reliable operation.

Shutdown Message

The following is the datalog shutdown message:

{-1} : alarm ["PinName", SiteNum] DVI:0020 : THERMAL SHUTDOWN. DCVI power dissipation limit has been exceeded.

where:

PinName	Name of the pin the shutdown occurred on.
SiteNum	Site number the shutdown occurred on.

Common Causes

A thermal shutdown can be caused by the following:

- Exceeding maximum power dissipation specification
- Shorted device
- Ambient temperature exceeding system specifications
- Shorted DUT socket or probe card

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DC75 Instrument Board : DC75 Shutdowns : DCTime Shutdown Conditions

DCTime Shutdown Conditions

A DCTime invokes a system shutdown when the following condition exists:

- Over Voltage

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DC75 Instrument Board : DC75 Shutdowns : Over Voltage

Over Voltage

The DCTime input voltage has exceeded the programmed front end voltage range.

Shutdown Message

The following are the datalog shutdown messages:

{-1} : alarm ["PinName", SiteNum] TMU:0015 : POSITIVE OVER VOLTAGE SHUTDOWN. Input voltage has exceeded the DCTime front end voltage range.

{-1} : alarm ["PinName", SiteNum] TMU:0016 : NEGATIVE OVER VOLTAGE SHUTDOWN. Input voltage has exceeded the DCTime front end voltage range.

where:

PinName	Name of the pin the shutdown occurred on.
SiteNum	Site number the shutdown occurred on.

Common Causes

An over voltage shutdown can be caused by the following:

- Defective device
- Incorrect front end range selection

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DC75 Instrument Board : DC75 Shutdowns : Clearing a Shutdown Condition

Clearing a Shutdown Condition

To clear a system shutdown, IG-XL must call an initialize routine. This means that:

- The test program in which the shutdown occurred must be stopped.
- DataTool must be stopped and started.
- The job must be revalidated.

Following a shutdown clear, make sure to understand and resolve the cause of the shutdown before rerunning the program.

For more information, see Shutdowns.

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DC75 Instrument Board : DC75 Software Licensing Requirements

DC75 Software Licensing Requirements

Operation of the DC75 does not require a software license.

For more information, see IG-XL Licensing in IG-XL Administration help.

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